Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech. | *Semester: IV* | *Time: 03 Hours*
Code: 101401 | *Subject: Civil Engineering Societal and Global Impact* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct answer of the following (Any seven questions only):* **[2 x 7 = 14]**

(a) The first Human Development report was published by the United Nations Development Programme (UNDP) in the year.....
(i) 1905 (ii) 1945
(iii) 1990 (iv) 1975

(b) The process of producing energy by utilizing heat trapped inside the earth surface is called
(i) Hydrothermal energy (ii) Geo-thermal energy
(iii) Solar energy (iv) Wave energy

(c) LEED means
(i) Leadership in Energy and Efficiency Design
(ii) Leadership in Energy and Efficiency Document
(iii) Leadership in Energy and Environmental Design
(iv) Leadership in Energy and Environmental Document

(d) Which of the following is not a Green house gas?
(i) Methane (ii) Hydrogen
(iii) Nitrous Oxide (iv) Ozone

(e) The type of coiling in DNA is
(i) Zig - Zag (ii) Left- handed
(iii) Opposite (iv) Right- handed

(f) Totipotency refers to
(i) Development of fruits from flowers in a culture
(ii) Development of an organ from a cell in a culture medium
(iii) Flowering in a culture medium
(iv) All of the above

(g) In biostatics, group of individuals taken for study is called as
(i) block (ii) population
(iii) group (iv) flock

(h) With reference to 'Hyperloop', which of the following statement is/are correct?
(I) Hyperloop is a futuristic transportation concept that intends to transport people to their destinations at super speeds
(II) Frictional losses and high maintenance costs are the main impediments in implementing hyperloop technology. select the correct answer using the code given below
(i) Only I (ii) Only II
(iii) Both I and II (iv) None of the above

(i) Floods can be prevented by
(i) afforestation (ii) cutting the forest
(iii) tilling the land (iv) removing the top soil

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(j) Which ISO 14000 series standard refers to the guidelines on Environmental Performance Evaluation?

(i) ISO 14001 (ii) ISO 14004

(iii) ISO 14010 (iv) ISO 14031

Q.2 What do you mean by tissue culture? Write difference between animal and plant tissue culture. Discuss in detail the use of tissue culture in agriculture and pharmaceutical industry. [14]

Q.3 (a) Write a note on India's ecological footprint. [7]

(b) What are the salient features of Environmental protection Act, 1986 [7]

Q.4 (a) What were socio-economic effects of the Industrial Revolution? Explain in detail. [7]

(b) What are the techniques adopted for reduction of green house emission in civil engineering projects. [7]

Q.5 (a) What are the methods to ensure sustainability. [7]

(b) Discuss the role of civil Engineering in shaping the world. [7]

Q.6 (a) Entist the modern engineering marvels of India. Describe any one in details. [7]

(b) Write short notes on [3.5×2=7]

(a) Agricultural Revolution

(b) Water Provisioning

Q.7 (a) What are the modes of transportation in India. Explain any on futuristic infrastructure project for transportation. [7]

(b) List seven measures to be taken as a civil Engineering for improving the quality of stakeholders. [7]

Q.8 List various non conventional sources of energy. Explain any four in detail emphasising two potential and limitations. [14]

Q.9 Write a note on any two of the following:- [7×2=14]

(a) Impact and causes of global warming

(b) Environmental metrics and monitoring

(c) Rehabilitation of Heritage structures with examples.

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